HELMHOLTZ AI Artificial Intelligence Cooperation Unit

HZDR

Abstracts-Explorer: LLM-Powered Semantic Exploration of Scientific Conference Proceedings

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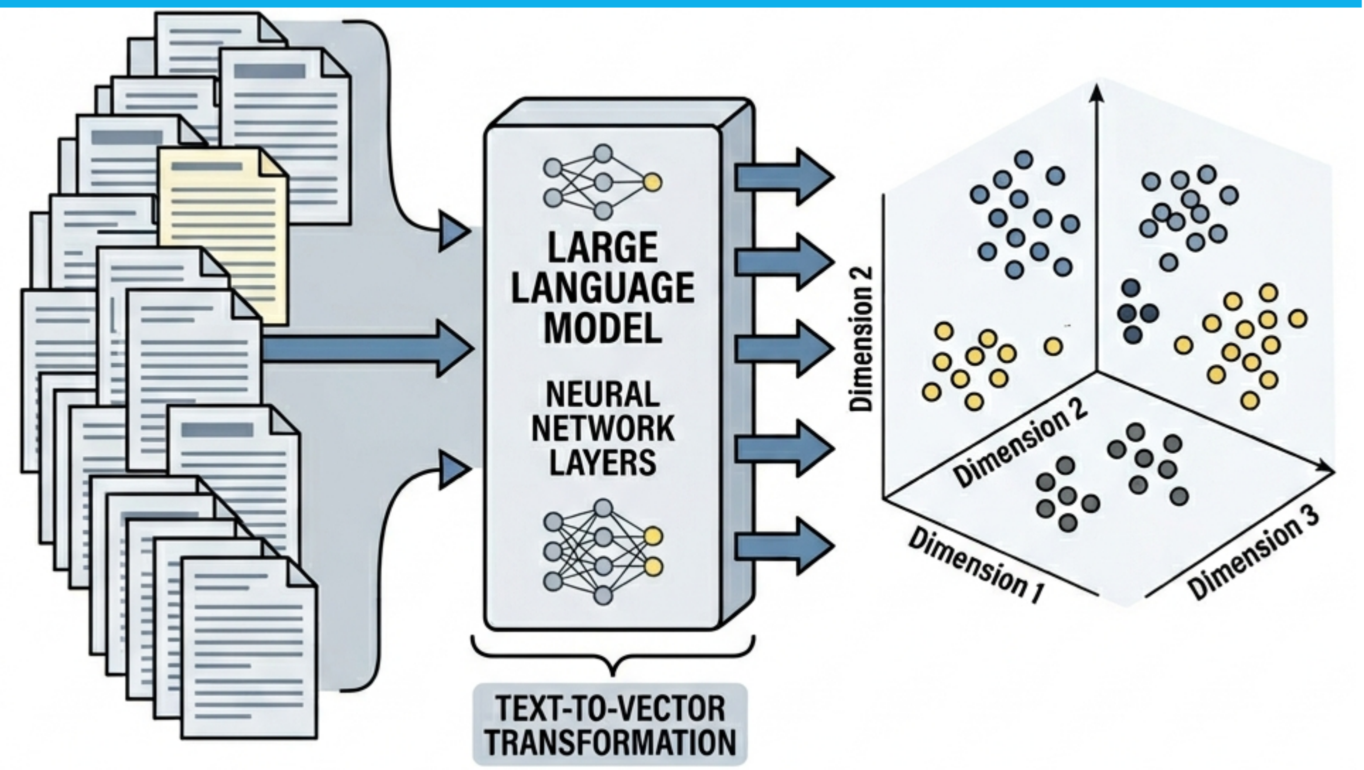
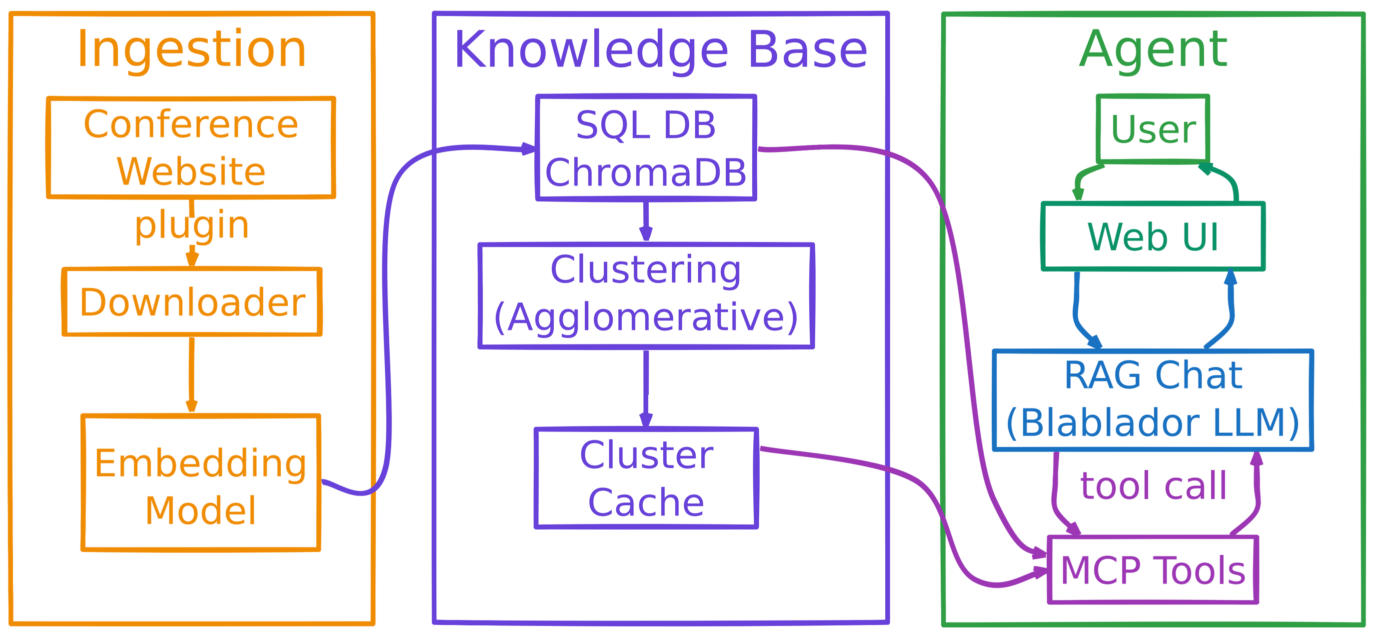
GitHub

PREMISE

You open the conference programme: 6000 abstracts. 10+ parallel tracks. You have an 8h flight to prepare. Where do you start?

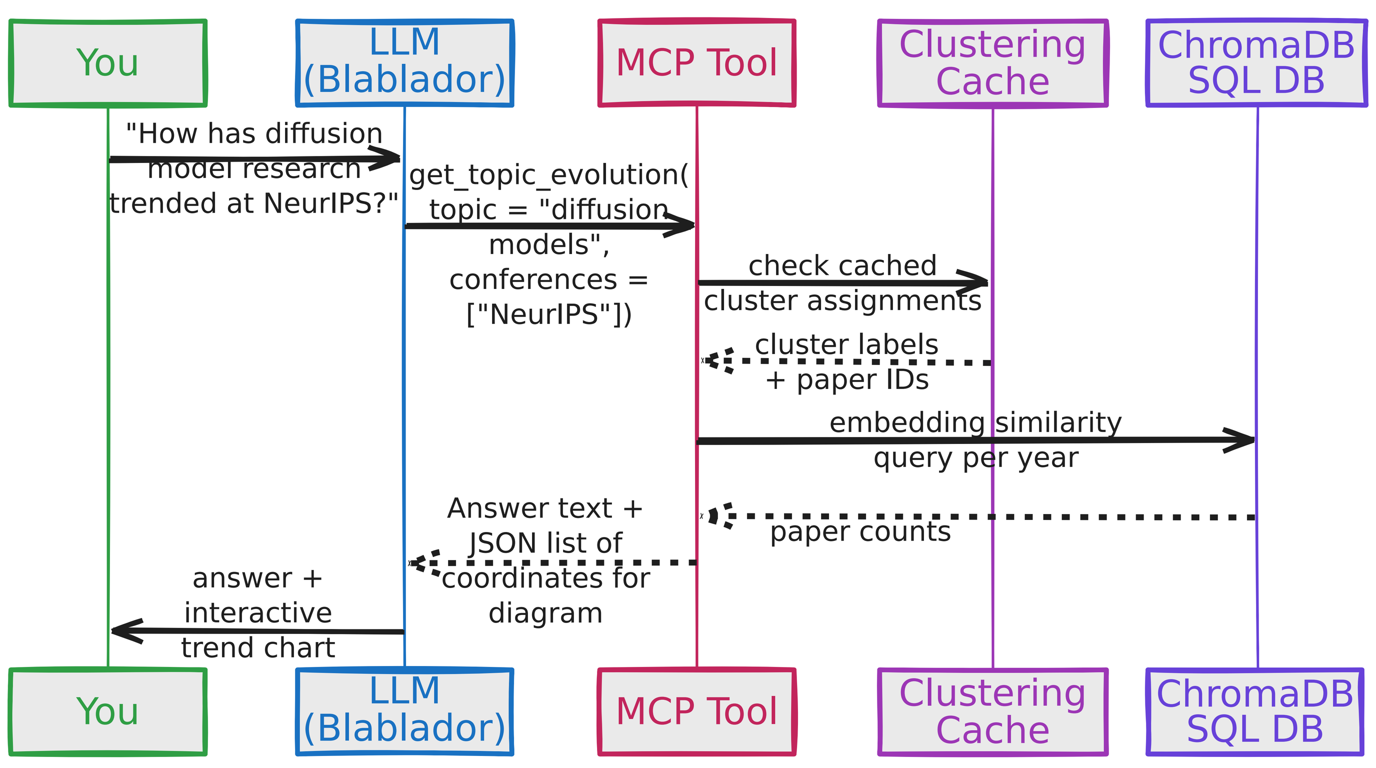


ARCHITECTURE

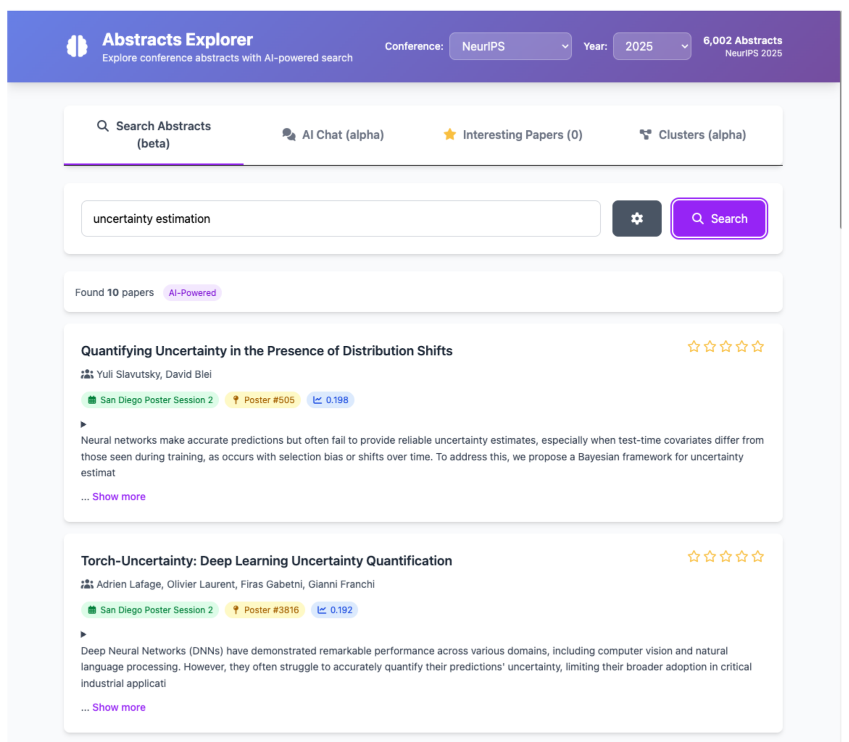


MCP TOOLS

Tool	Example trigger phrase	Returns
`get_conference_topics`	"What are the main themes at NeurIPS 2025?"	cluster names, keywords, paper counts
`analyze_topic_relevance`	"How prominent is uncertainty quantification?"	paper count within embedding radius
`get_topic_evolution`	"How has deep learning changed since 2020?"	per-year paper counts → Plotly chart
`search_papers`	"Find papers about graph neural networks"	ranked paper list with titles & abstracts
`get_cluster_visualization`	"Show me the topic landscape"	2-D scatter data → Plotly chart
`get_paper_details`	"Who wrote that paper? Where is the poster?"	authors, PDF, session, room, awards



SEMANTIC SEARCH

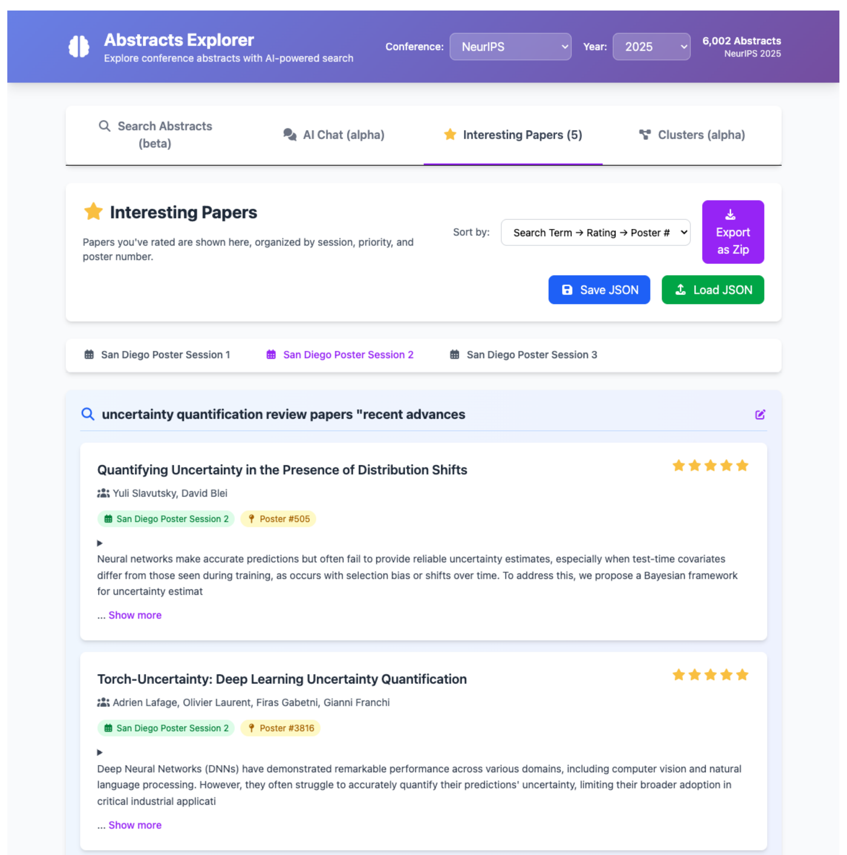


Find relevant abstracts based on **meaning** rather than keywords

LLM-ranked results with related topic suggestions

Filters: year, conference, track, event type

INTERESTING PAPERS



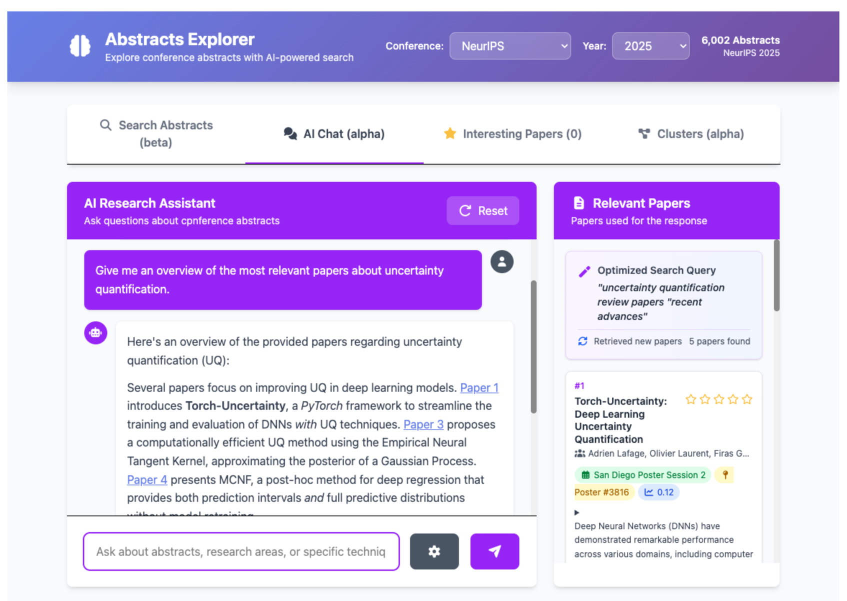
Curate your personal schedule

Rate abstracts 1–5 stars while browsing

Organised by session, priority, poster number

Export to markdown conference notes

CHAT

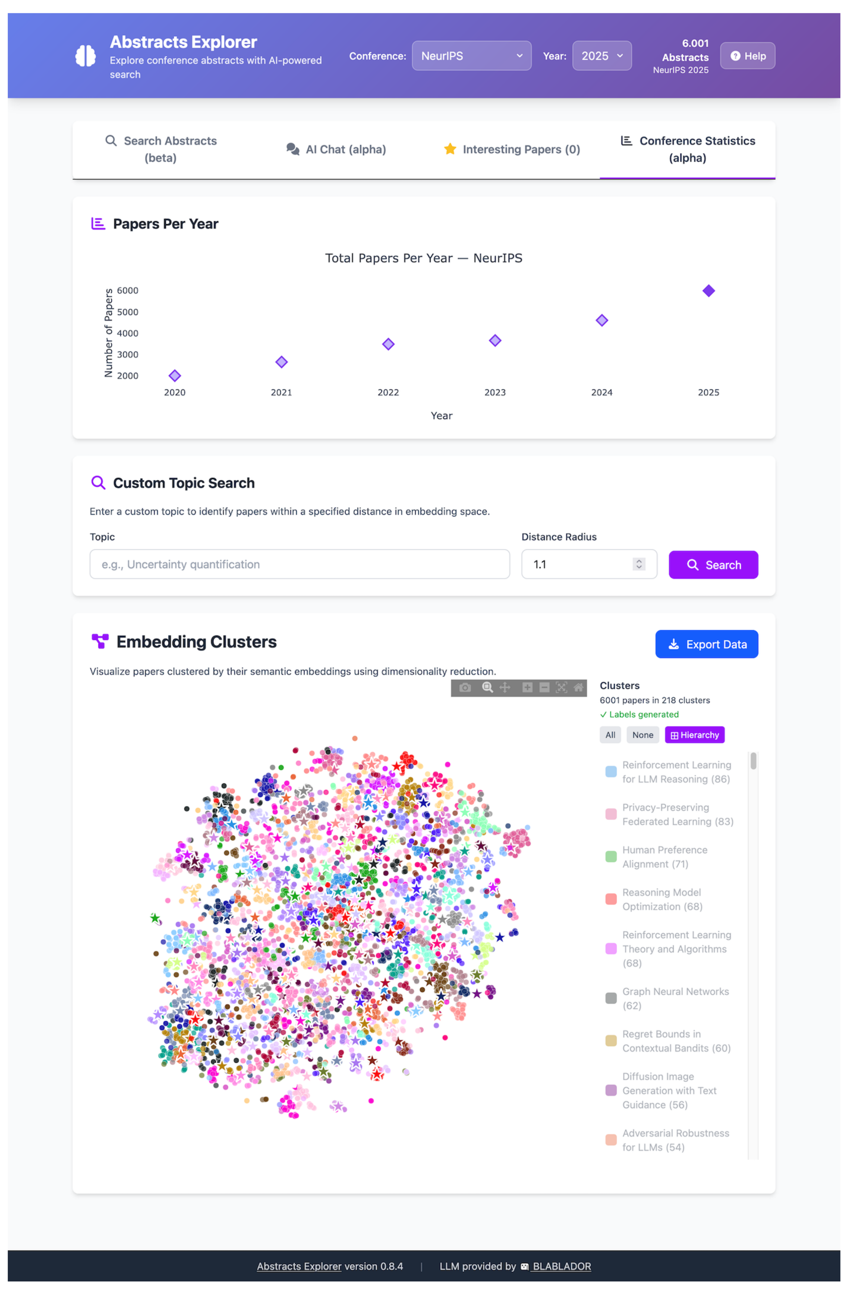


Ask questions about the conference

Conversational RAG interface with query rewriting

MCP tools for topic trends and cluster analysis

CONFERENCE STATISTICS



Explore the topic landscape

Papers per year trend chart

Search for topics and see how they relate to other topics and how they evolved over time

Interactive cluster visualisation

Topic trend over the years (after search)